

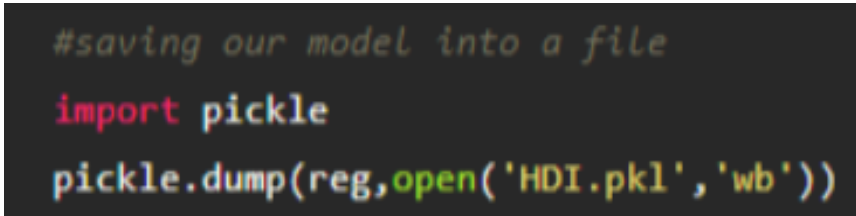
Saving the Trained Model Using Pickle

After successfully training and validating the machine learning model, the trained model is saved for future use by serializing it into a **.pkl** (Pickle) file. Saving the model eliminates the need to retrain it whenever predictions are required, thereby saving time and computational resources.

In the deployed Flask web application, the saved model is loaded directly from the Pickle file and used to predict the Human Development Index (HDI) for new user inputs.

Pickle is a built-in Python module that supports the serialization and deserialization of Python objects. Serialization converts a Python object into a byte stream that can be stored on disk, while deserialization reconstructs the object from the stored file. Since machine learning models are Python objects, Pickle provides an efficient and reliable mechanism for saving and loading trained models for deployment.

Code:



```
#saving our model into a file  
import pickle  
pickle.dump(reg, open('HDI.pkl', 'wb'))
```

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```